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Bio-Inspired Hybrid Dielectric for Capacitive and Triboelectric Tactile Sensors with High Sensitivity and Ultrawide Linearity Range

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The trade-off between sensitivity and linearity is critical for preserving the high pressure-resolution over a broad range and simplifying the signal processing/conversion of flexible tactile sensors. Conventional dielectrics suffer from the difficulty of quantitatively controlling the interacted mechanical and dielectric properties, thus causing the restricted sensitivity and linearity of capacitive sensors. Herein, inspired by human skin, a novel hybrid dielectric composed of a low-permittivity (low-*k*) micro-cilia array, a high-permittivity (high-*k*) rough surface, and micro-dome array is developed. The pressure-induced series-parallel conversion between the low-*k* and high-*k* components of the hybrid dielectric enables the linear effective dielectric constant and controllable initial/resultant capacitance. The gradient compressibility of the hybrid dielectric enables the linear behavior of elastic modulus with pressures, which derives the capacitance variation determined by the effective dielectric constant. Therefore, an ultrawide linearity range up to 1000 kPa and a high sensitivity of 0.314 kPa^{-1} are simultaneously achieved by the optimized hybrid dielectric. The design is also applicable for triboelectric tactile sensors, which realizes the similar linear behavior of output voltage and enhanced sensitivity. With the high pressure-resolution across a broad range, potential applications such as healthcare monitoring in diverse scenarios and control command conversion via a single sensor are demonstrated.

1. Introduction

Flexible tactile sensors that can directly transfer the tactile information into electric signals have attracted wide attention because of their potential applications in healthcare monitoring, human-machine interfaces, intelligent control, Internet of things (IoT), etc.[1–6] Among different sensor types, dielectric based sensors such as capacitive and triboelectric tactile sensor are more attractive taking advantages of the simple structure, facile signal acquisition, low energy consumption, etc.[7–9] To adapt multifunctional application scenarios, the desired sensor devices should have the high sensitivity and wide linear response range. The high sensitivity enables the precise recognition of tiny pressure variations with high signal-to-noise ratio.[10,11] The wide linearity range can preserve the high-pressure resolution across a broad pressure spectrum and simplify the data processing and conversion.[12,13] Conventional capacitive tactile sensors with the solid dielectric sandwiched between electrodes suffer from the restricted sensitivity due to the low compressibility of the solid dielectric.[12,14] Designing the micro-structured dielectric with higher compressibility has been proven as an effective approach to improve the sensitivity.[15–18] To date, various microstructures such as micro-pillars, micro-domes, micro-pyramids, and micro-cilia, etc. and multi-layer structures have been extensively studied.[15,19–22] Such microstructures, however, mainly take effect in low-pressure region, thus leading to the narrow linearity range and dramatically reduced pressure resolution with increasing pressures. Although increasing the dielectric constant of the micro-structured dielectric with high-permittivity particles (e.g., $CaCu_3Ti_4O_{12}$, $BaTiO_3$, etc.) or conductive fillers (e.g., carbon nanotube (CNT), MXene, etc.) can further enhance the sensitivity, the trade-off between sensitivity and linearity (e.g., a high sensitivity over 0.1 kPa^{-1} in a linearity range up to 1 MPa) is yet to be fully resolved.[23–25] Moreover, with different micro-structured dielectrics, the linearity range will be dramatically varied due to the interacted mechanical and dielectric properties. To the best of our knowledge, it is still not clear how exactly the ultrawide linearity range of capacitive tactile sensor can be realized even though continuous efforts have been applied to improve the sensing sensitivity during the past years. Similar challenges also exist in triboelectric tactile sensors. Although extensive triboelectrification materials (e.g., polydimethylsiloxane (PDMS), poly(vinylidene fluoride),

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Nylon, poly(methyl methacrylate), etc.) and microstructures (e.g., fabrics, micro-domes, micro-pyramids, etc.) have been developed to improve the output voltage, the triboelectric tactile sensor that can exhibit an ultrawide linearity range is rarely reported.[26–33] It is still challenging to synergistically enhance the sensitivity and linearity of both capacitive and triboelectric tactile sensors with clear mechanism.

As a natural sensing tissue, the skin with hybrid structures can exhibit the high tactile resolution in a wide range of tactile stimuli. Inspired by the skin, we herein presented a novel hybrid dielectric comprised a low-permittivity (low-*k*) carbonyl iron powder/NdFeB/PDMS (CIP/NdFeB/PDMS) micro-cilia array (MCA), a high-permittivity (high-*k*) CNT/PDMS rough surface and micro-dome array (RDA) as the active layer of capacitive tactile sensors. Different from conventional dielectrics, the hybrid dielectric with combined low-*k* MCA and high-*k* RDA exhibited the gradient compressibility and dielectric property to effectively regulate the capacitance variation. Firstly, the low-*k* and high-*k* components of the hybrid dielectric could be converted from series to parallel connection (series-parallel conversion) during the pressure-induced mechanical deformations. Such a unique series-parallel conversion, based on the rational collocation of hybrid components, enabled the linearly varied effective dielectric constant of the hybrid dielectric and the flexibly regulated initial/resultant capacitance of the sensor. Secondly, the gradient compressibility of the hybrid dielectric contributed to the linear behavior of elastic modulus with pressures. With the linear elastic modulus, the linearity of capacitance variation will be mainly determined by the change in effective dielectric constant of the hybrid dielectric. As a result, the ultrawide linearity range up to 1000 kPa with excellent linearity (R^2 = 0.99) and the high sensitivity of 0.314 kPa^{-1} were simultaneously realized, which is first reported. We also demonstrated that our design of hybrid dielectric was applicable for triboelectric tactile sensors. The combined low-*k* MCA and high-*k* RDA with linear effective dielectric constant not only enhanced the triboelectrification and electrostatic induction for output voltage amplification but also realized the linear relationship between output voltage and applied pressures. As a result, the triboelectric tactile sensor with the ultrawide linearity range (up to 1000 kPa) and obviously enhanced sensitivity was obtained. As a proof-of-concept, both the proposed capacitive and triboelectric tactile sensor were successfully demonstrated as a healthcare monitor to detect the physiological signals such as the artery pulse, respiratory rate, joint flexion, walking and running status, etc. Moreover, we also demonstrated the potential of the sensors as a transmitter to deliver configurable commands via combinations of forces with different intensities. Attributed to the high-pressure resolution across the ultrawide pressure spectrum, the sensor can generate different signal code groups with combinatorial force inputs in a single sensor device, which avoids the high demand of numerous sensor pixels and complex signal processing for multiple signal communications. With the dramatically improved sensing capabilities, we believe that the novel design of hybrid dielectric can provide a promising strategy to develop high-performance wearable devices for applications in healthcare monitoring, human-machine interfaces, etc. in the near future.

2. Results and Discussion

2.1. Design Principle of Skin-Inspired Hybrid Dielectric

The skin, which can perceive and transmit vital biological signals from the inner organs, blood vessels, muscles, and dermis/epidermis, is the largest organ of the integumentary system in human body.[34] As the outermost layer of skin, the hybrid epidermis composed of the hair shaft, corneal layer and basal layer possesses the gradient compressibility to generate different mechanical deformations under external tactile stimuli (**Figure 1**a). These mechanical deformations can be transmitted to the mechanoreceptors, which induce the variation in biopotentials corresponding to the tactile intensity. As a result, the skin can perceive a wide range of external tactile stimuli with a high resolution. Inspired by the structure of skin, we herein designed a novel hybrid dielectric composed of a low-*k* MCA, a high-*k* RDA as the active layer for capacitive tactile sensors (Figure 1b). The high-*k* RDA pattern was fabricated with the CNT/PDMS mixture, and the low-*k* MCA was obtained by spraying (CIP/NdFeB/PDMS) magnetic composite on the RDA assisted by external magnetic field (see the Experimental Section and Figure S1 in the Supporting Information for the detailed fabrication methodology). The CIP/NdFeB/PDMS with unchanged content of CIP and NdFeB particles exhibited the fixed dielectric constant of ≈25 in this work (at 1 kHz), while the dielectric constant of CNT/PDMS was dependent on the content of CNT fillers (Figure S2, Supporting Information). The maximum dielectric constant of ≈2100 was obtained with the CNT content of 3.5 wt% in PDMS (at 1 kHz). The optical and scanning electron microscopy (SEM) images of the typical hybrid dielectric are presented in Figure 1c (detailed dimensions, size regulations, and characterizations of the hybrid dielectric can be referred to Figures S3–S8 in the Supporting Information).

Similar to the human skin, the hybrid dielectric also exhibits the gradient compressibility. The hair-inspired MCA with excellent flexibility and compressibility can mechanically respond to subtle pressures with obvious deformation.[20,21] The corneal layer and basal layer inspired RDA will respond to increasing pressures with dramatically decreased deformations because of the relatively larger scales and lower compressibility. Such a unique deformation behavior enables the low-*k* MCA and high-*k* RDA to take effect in different pressure regions. Specifically, during the mechanical deformation, the low-*k* and high-*k* components can be converted from series to parallel connection (i.e., series-parallel conversion), which further contributes to the gradient dielectric property of the hybrid dielectric. These distinctive features derive the customizable mechanical and dielectric behavior of the hybrid dielectric to realize the linear capacitance variation. Firstly, attributed to the series-parallel conversion, the dielectric behavior of the hybrid dielectric can be customized by regulating the dielectric behavior of low-*k* and high-*k* components separately. With appropriate design of the structural collocation and intrinsic dielectric constant of the low-*k* MCA and high-*k* RDA, the linear variation in effective dielectric constant of the hybrid dielectric can be realized (Figure 1d). Secondly, the gradient compressibility of the hybrid dielectric contributes to the

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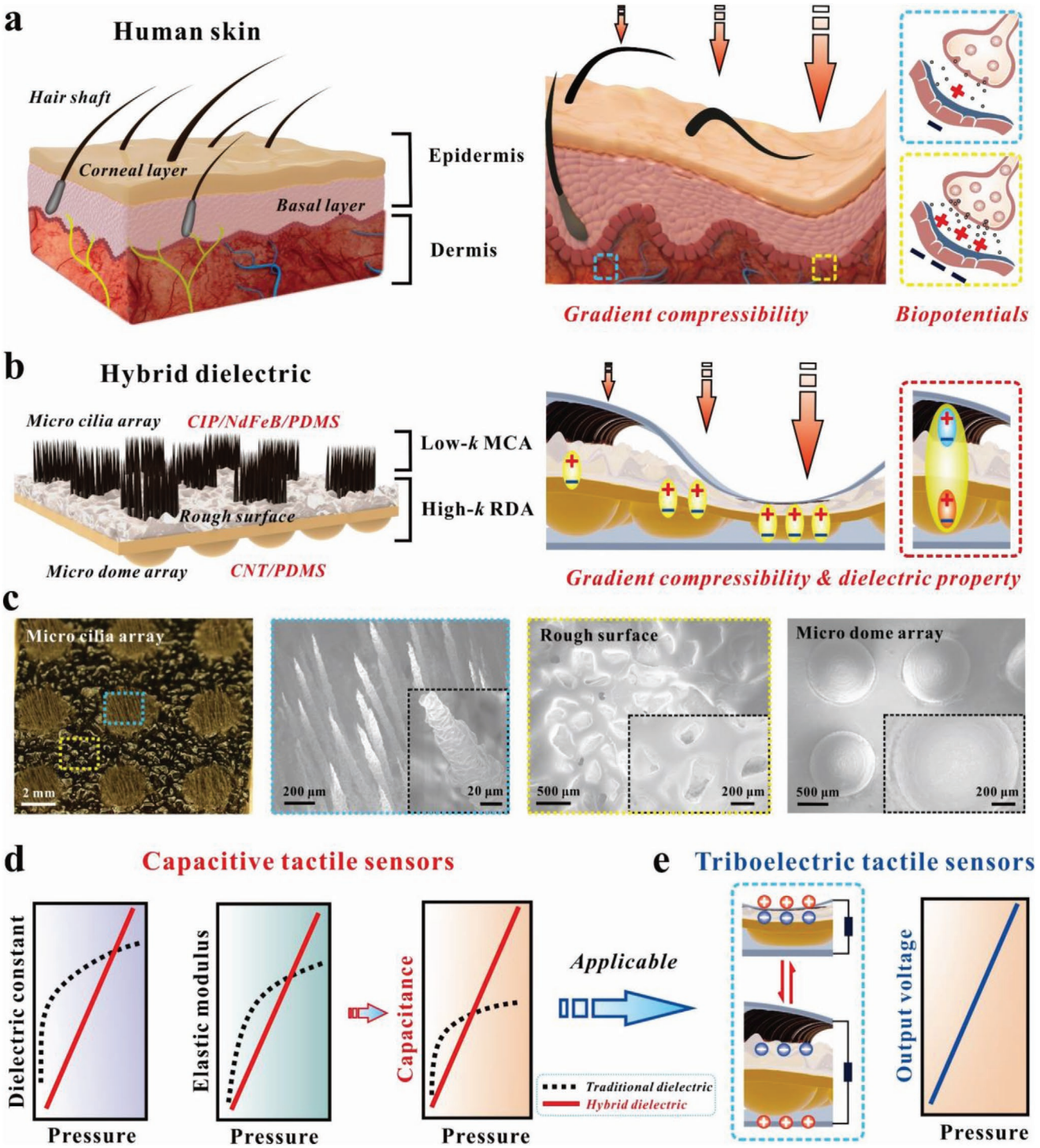

Figure 1. Schematic illustration of the structure and features of a) human skin and b) bio-inspired hybrid dielectric. c) Typical optical and SEM images of the hybrid dielectric. d) Customizable mechanical and dielectric behavior of the hybrid dielectric for capacitive tactile sensor with ultrawide linearity range. e) Applicability of the hybrid dielectric for triboelectric tactile sensors with ultrawide linearity range.

linear behavior of elastic modulus with the applied pressure. With the linear elastic modulus, the linearity of capacitance variation will be mainly determined by the dielectric behavior of the hybrid dielectric. Note that the regulation of hybrid components to customize the dielectric behavior will not affect the gradient compressibility, the linear elastic modulus and hence the dependence of capacitance linearity on dielectric behavior can be maintained. With the linear effective dielectric constant of the hybrid dielectric, the linear capacitance variation can therefore be realized. Moreover, as the low-k

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MCA and high-*k* RDA take effect under different pressure intensities, the initial and resultant capacitance of the sensor can also be flexibly regulated to realize an enhanced sensitivity. The detailed principle will be discussed in the following section. In comparison, conventional dielectrics suffer from the interacted, nonlinear mechanical and dielectric behaviors, thus imposing the difficulty to simultaneously address the linearity and sensitivity. In addition to the capacitance, the contact electrification and electrostatic effects between the hybrid dielectric and electrodes will also be affected by the concomitant mechanical deformation and dielectric behavior. Therefore, our design is also applicable for self-powered triboelectric tactile sensors which can linearly convert mechanical stimuli into electrical signals for perceiving the dynamic tactile information (Figure 1e).

2.2. Customization of Dielectric and Mechanical Behavior of Hybrid Dielectric for Capacitive Tactile Sensors

Figure 2a depicts the principle of the series-parallel conversion for customizing the dielectric behavior of the hybrid dielectric. As mentioned above, the mechanical deformation of low-*k* MCA and high-*k* RDA mainly occurs in the low and high-pressure region, respectively. Under low pressures, the top electrode and the high-*k* RDA are separated by the low-*k* MCA, thus resulting in the series connection of the low-*k* MCA and high-*k* RDA at this stage (i.e., SC stage). However, with increasing pressures, the MCA will be fully compressed, and the high-*k* RDA can directly contact with both the top and bottom electrodes. Then the hybrid dielectric becomes the series connection of the MCA and the RDA components covered underneath

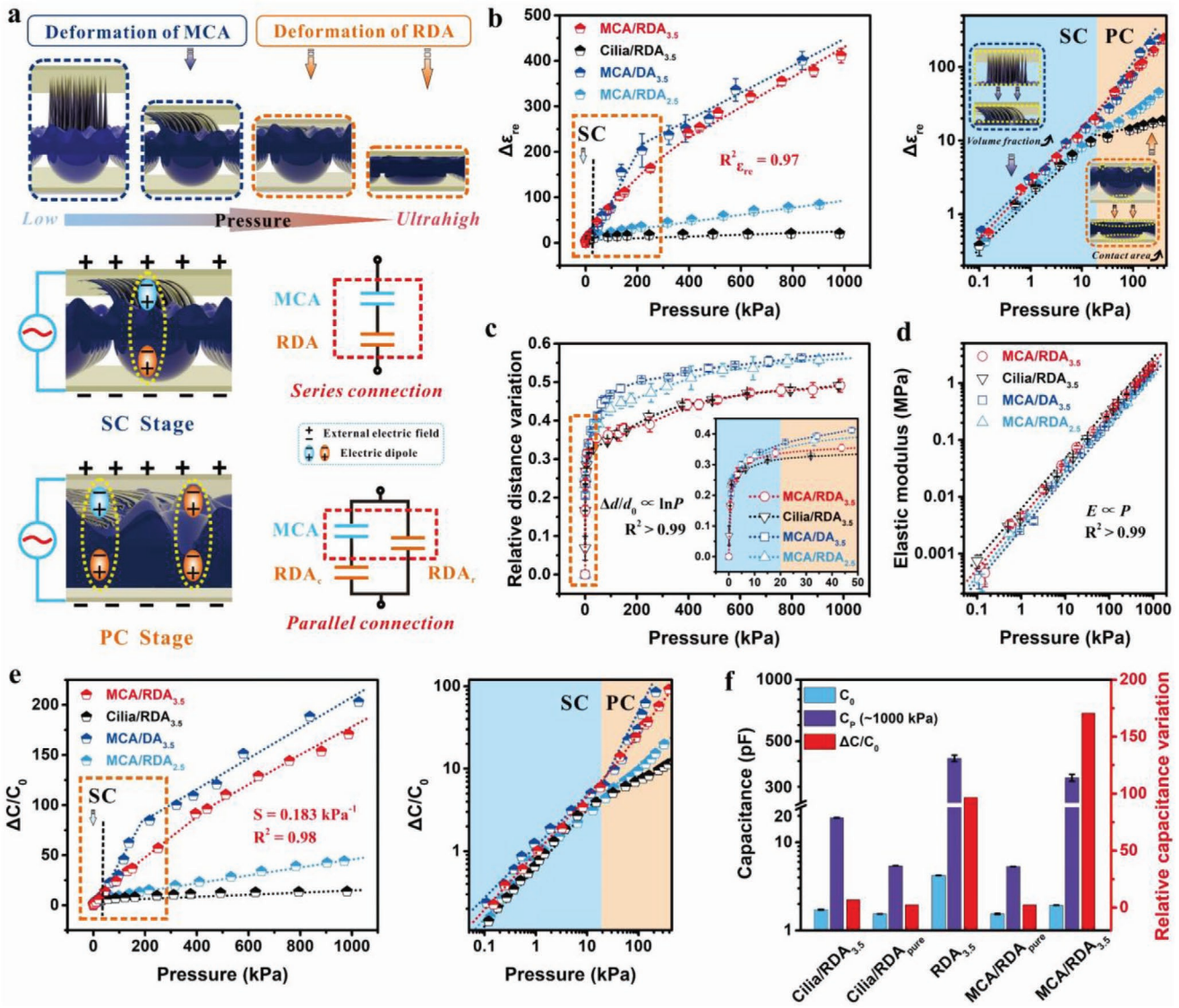

Figure 2. a) Principle of series-parallel conversion of the hybrid dielectric. b) Variation in effective dielectric constant of hybrid dielectrics with different combinations of low-*k* and high-*k* components. c) Relative distance variation of the mentioned hybrid dielectrics. d) Elastic modulus of the mentioned hybrid dielectrics. e) Relative capacitance variation of sensors based on the mentioned hybrid dielectrics. f) Initial capacitance without loading pressures, resultant capacitance under ≈1000 kPa and the relevant relative capacitance variation of the sensors based on different dielectrics.

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(RDA_c) in parallel with the remained RDA components (RDA_r). The series connection will dramatically restrict the impact of covered components (RDA_c), and the content of remained components (RDA_r) is much greater than that of the covered one (RDA_c) in this work. Consequently, the hybrid dielectric can be approximately regarded as the parallel connection of the low-k MCA and high-k RDA at this stage (i.e., PC stage). From this perspective, the low-k MCA and high-k RDA can be converted from series to parallel connection (i.e., series-parallel conversion) during the pressure-induced mechanical deformation. Such a unique series-parallel conversion enables the low-k MCA and high-k RDA to take effect on the dielectric behavior of hybrid dielectric in different pressure regions. The low-k MCA will dominate the dielectric behavior of the hybrid dielectric at SC stage (low-pressure region) in virtue of the series connection. The high-k RDA will dominate the dielectric behavior at PC stage (high-pressure region) because of the parallel connection. Then the global dielectric behavior of the hybrid dielectric can be customized by regulating the dielectric behavior of low-k and high-k components separately.

Herein, we firstly examined the dielectric behavior of hybrid dielectrics with similar low-k MCA but different high-k components, e.g., MCA/$RDA_{3.5}$, MCA/$DA_{3.5}$ and MCA/$RDA_{2.5}$, as shown in Figure 2b. The dimensions of MCA, RDA, and DA were maintained unchanged in different hybrid dielectrics (detailed dimensions can be referred to Table S1 in the Supporting Information). "DA" means the smooth surface was introduced on the backside of micro-dome array. The subscript "2.5" and "3.5" represent the employed CNT content in PDMS (2.5 and 3.5 wt%) for the high-k components. It can be observed that the MCA/$RDA_{3.5}$ hybrid dielectric exhibited the approximately linear variation in effective dielectric constant (ε_{re}) over the full scale of pressure spectrum ($R^2 = 0.97$, 0–1000 kPa). In comparison, both the MCA/$DA_{3.5}$ and MCA/$RDA_{2.5}$ exhibited the segmented variation curve of ε_{re} (Figure 2b; the measurement of ε_{re} can be referred to the Experimental Section). These different ε_{re} variations indicate that the dielectric behavior of hybrid dielectrics can be customized by regulating the high-k components separately attributed to the series-parallel conversion. Moreover, it was observed that the mentioned dielectrics exhibited the similar linear variation in ε_{re} in the range of 0–20 kPa but dramatically varied variations in ε_{re} in the range of ≈20–1000 kPa. Such results indicate that the boundary between SC and PC stage of the hybrid dielectrics exists at ≈20 kPa, which can also be confirmed by the optical images that show the dynamic deformation of the mentioned dielectrics under different pressures (Figure S9, Supporting Information). At SC stage (0–20 kPa), the variation in ε_{re} of the hybrid dielectric is mainly determined by the volume fraction of the low-k MCA during compression, as indicated by the inset of Figure 2b. Theoretical analysis in Section S1 in the Supporting Information reveals that the compression of MCA will increase the volume fraction of low-k components and result in the linearly increased ε_{re} with pressures. Without changing the low-k MCA, the mentioned dielectrics thus exhibited the similar linear variation in ε_{re} at SC stage. However, when it turns to PC stage (≈20–1000 kPa), the variation tendency of ε_{re} is mainly determined by the contact area between the electrodes and high-k components (inset of Figure 2b). In principle, the variation in contact area will be affected by both the roughness of the rough surface and the dimension of micro-domes of the high-k components. Here we will mainly discuss the influence of rough surface because the micro-domes with slightly varied structural parameters exhibited the similar performance (Figure S10, Supporting Information). With the higher roughness (≈350 μm), the RDA based high-k components could exhibit the continuous variation in contact area, which in turn contributed to the approximately linear variation in ε_{re} of the MCA/$RDA_{3.5}$ and MCA/$RDA_{2.5}$ at PC stage. Such statements can also be proved by the theoretical analysis in Section S2 and the fitting result in Figure S11 in the Supporting Information. In comparison, the MCA/$DA_{3.5}$ without the rough surface exhibited the much larger contact area between high-k components and electrodes at the beginning of PC stage, thus leading to the segmented variation in ε_{re} at PC stage. Similar nonlinear dielectric behavior could also be observed in the MCA/RDA based hybrid dielectric with a smaller degree of surface roughness (≈100 μm), as shown in Figure S12 in the Supporting Information. These results reveal that the rough surface of the high-k components play a significant role in the linearity of dielectric behavior. Moreover, compared with the MCA/$RDA_{3.5}$, the MCA/$RDA_{2.5}$ with a lower CNT content (2.5 wt%) in RDA components exhibited the segmented variation curve of ε_{re} at the boundary between SC and PC stage. Such results can be explained by the decreased CNT content and hence lower intrinsic dielectric constant of the high-k components in MCA/$RDA_{2.5}$, which renders a relatively lower increase rate of ε_{re} at PC stage than that at SC stage. Similar nonlinear dielectric behavior could also be observed in the MCA/RDA based hybrid dielectric without the CNT fillers (i.e., with much lower intrinsic dielectric constant) in the RDA components (Figure S12, Supporting Information). Such results reveal that the intrinsic dielectric constant of the high-k components is also critical to the linearity of dielectric behavior. We also experimentally examined the dielectric behavior of the Cilia/$RDA_{3.5}$ dielectric with fully covered micro-cilia and unchanged high-k $RDA_{3.5}$ components for comparison (Figure 2b). Due to the full coverage of micro-cilia, the Cilia/$RDA_{3.5}$ suffered from the monotonous series connection of low-k and high-k components over the entire pressure spectrum, which dramatically restricted the variation in ε_{re} at PC stage. These results sufficiently indicate that the series-parallel conversion enables the customizable dielectric behavior of hybrid dielectrics by regulating the collocation of low-k and high-k components. Based on the appropriate collocation (e.g., MCA/$RDA_{3.5}$), the linear dielectric behavior of the hybrid dielectric over the full scale of pressure can be realized.

Besides the dielectric behavior, the mechanical deformation is another important parameter to determine the linearity of capacitance variation. Herein, we also experimentally examined the mechanical deformation behavior of the dielectrics (MCA/$RDA_{3.5}$, MCA/$DA_{3.5}$, MCA/$RDA_{2.5}$, and Cilia/$RDA_{3.5}$) under different pressures, as shown in Figure 2c. It can be observed that all the dielectrics exhibited the natural logarithm relationship between the relative distance variation and the applied pressure ($\Delta d/d_0 \propto \ln P$, $R^2 > 0.99$), regardless of the different collocations of low-k and high-k components (the detailed expressions of $\Delta d/d_0$ can be referred to Table S2, Supporting Information). Such a result can be attributed to the gradient compressibility

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of the hybrid dielectrics which enables the high deformation in low-pressure region and the low deformation in high-pressure region. Moreover, the inset indicates that the relative distance variation of the dielectrics dramatically decreased after ≈20 kPa, which is also consistent with the boundary between SC and PC stage discussed above (Figure 2b). With the natural logarithm relationship ($\Delta d/d_0 \propto \ln P$), the linear elastic modulus of the mentioned dielectrics with pressures ($E \propto P$, $R^2 > 0.99$) can be obtained, as shown in Figure 2d (the detailed measurement of elastic modulus can be referred to the Experimental Section). The theoretical analysis in Section S3 and the fitting results in Figure S13 in the Supporting Information further confirm that the natural logarithm relationship between the mechanical deformation and the applied pressure can derive the approximately linear behavior of elastic modulus for our hybrid dielectric. The similar natural logarithm relationship and linear elastic modulus could also be observed in other MCA/RDA based hybrid dielectrics (Figure S14, the Supporting Information). These results support that the customization of dielectric behavior via regulating the hybrid structure (e.g., micro-cilia, MCA, surface roughness, etc.) will not affect the mechanical behavior of the hybrid dielectrics thanks to the gradient compressibility. From this perspective, our design of hybrid dielectric allows the separate optimization on the dielectric and mechanical behaviors toward the sensing linearity.

As the distance between the facing electrodes can also be expressed as $d = d_0 - Pd_0/E$ according to the definition of elastic modulus $\left(E = \frac{P}{\Delta d/d_0}\right)$, the capacitance can be rewritten as $C = \frac{\varepsilon_0 \varepsilon_{re} A/d_0}{1 - P/E}$, where ε_0 is the permittivity of vacuum and A the facing area between electrodes (constant in this work). With the approximately linear relationship between the elastic modulus and the applied pressure, the linearity of capacitance variation will be mainly determined by the effective dielectric constant (ε_{re}). As a result, the similar variation tendency between the relative capacitance variation and the effective dielectric constant was observed in capacitive sensors based on the mentioned dielectrics, as shown in Figure 2e and Figure S15 in the Supporting Information. With the approximately linear effective dielectric constant, the MCA/RDA$_{3.5}$ based sensor thus exhibited the ultrawide linearity range up to 1000 kPa with a high sensitivity of 0.183 kPa^{-1} (Figure 2e). Here, the sensitivity is defined by $S = \frac{\partial(\Delta C/C_0)}{\partial P}$, where C is the resultant capacitance under pressures, C_0 the initial capacitance and P the applied pressure.[11,17,21,35] We also systematically evaluated the influence of structural properties of MCA, e.g., height and coverage, on the sensing performance of the sensor without changing the RDA components (i.e., RDA$_{3.5}$), as shown in Figure S16 in the Supporting Information. The sensitivity could be enhanced with the increase of height or the decrease of coverage of the MCA, while the ultrawide linearity range was unchanged. The higher height and lower coverage of MCA contributes to the lower initial capacitance and higher resultant capacitance of the sensor, respectively, which can improve the sensitivity. However, the regulation of MCA will not change the linear behavior of elastic modulus and effective dielectric constant, which preserves the ultrawide linearity range of the sensor. We also noticed that the sensing performance would not be dramatically affected by the form of MCA with the certain height and coverage (Figure S16d, Supporting Information). For convenient investigation, we herein maintained the MCA with the parameters consistent with MCA/RDA$_{3.5}$ hybrid dielectric in the following study.

In addition to the sensor linearity, the hybrid dielectric with combined low-k MCA and high-k RDA (and hence the pressure-induced series-parallel conversion) also enables the flexible regulation of initial and resultant capacitance for sensitivity enhancement. Figure 2f presents the initial capacitance without loading pressures (C_0), the resultant capacitance under a pressure of ≈1000 kPa (C_P) and the relative capacitance variation ($\Delta C/C_0$) of different dielectrics (the detailed variation curves can be found in Figure S17 in the Supporting Information). The dimensions of the dielectric components were also maintained unchanged (Table S1, Supporting Information). It can be clearly observed that the MCA/RDA$_{3.5}$ hybrid dielectric with the combined low-k MCA and high-k RDA (and hence series-parallel conversion) simultaneously exhibited the low initial capacitance (C_0) and high resultant capacitance (C_P), thus contributing to the relatively higher $\Delta C/C_0$. In comparison, the MCA/RDA$_{pure}$ dielectric with monotonous low-k components suffered from the restricted C_P, and the individual RDA$_{3.5}$ dielectric with monotonous high-k components suffered from the relatively higher C_0 ("pure" indicates no CNT fillers). Therefore, they both exhibited the restricted $\Delta C/C_0$. Moreover, when the RDA components were fully covered by the micro-cilia (i.e., Cilia/RDA based dielectrics), the series connection of micro cilia and RDA components would take effect over the entire pressure spectrum, which in turn dramatically restricted the C_P (and hence $\Delta C/C_0$) regardless of the high-k components. Such statements of the sensitivity enhancement can also be proved by the simulation results in Figure S18 in the Supporting Information (the detailed simulation can be referred to Section S4 in the Supporting Information). These results indicate that with the combination of low-k MCA and high-k RDA, the pressure-induced series-parallel conversion can effectively improve the value of relative capacitance variation and hence the sensitivity.

2.3. Sensing Performance of Capacitive Tactile Sensors and Applicability of Hybrid Dielectric for Triboelectric Tactile Sensors

In principle, reducing the thickness of the solid CNT/PDMS layer in the high-k components of MCA/RDA$_{3.5}$ can further improve the sensitivity without affecting the sensor linearity. We herein examined the sensing performance of the sensor based on the MCA/RDA$_{opt}$ hybrid dielectric with a decreased thickness, as shown in **Figure 3**a (the detailed dimensions can be referred to Table S1 in the Supporting Information). The high sensitivity of 0.314 kPa^{-1} and the similar ultrawide linearity range up to 1000 kPa ($R^2 = 0.99$) were simultaneously obtained. With the optimized sensing performance, the MCA/RDA$_{opt}$ based sensor was selected for the following study. Taking advantages of the high sensitivity and ultrawide linearity range, the sensor can exhibit the high pressure-resolution across a broad pressure range. Therefore, the tiny pressure variations could be successfully recognized even when the low pressure (e.g., 1 and 10 kPa) or high pressure (e.g., 100 and 450 kPa) had been pre-loaded to the device, as presented in Figure 3b. We also examined the

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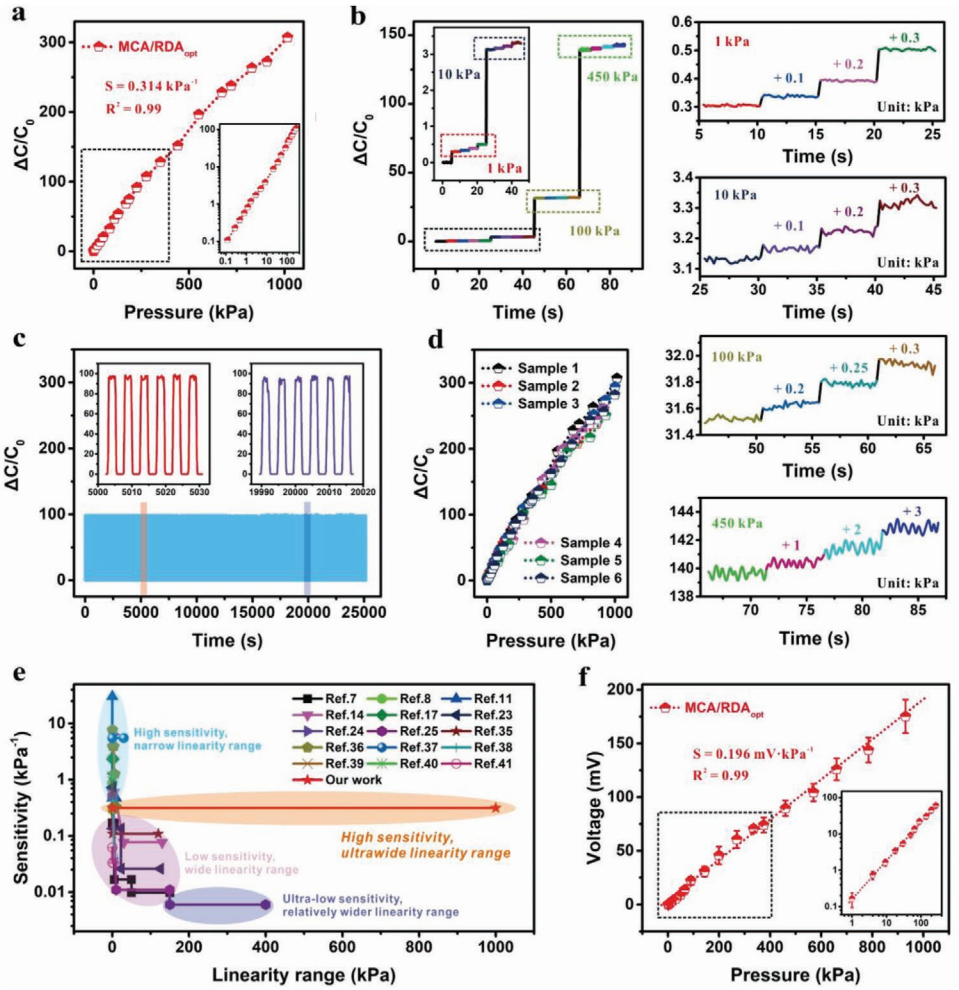

Figure 3. a) Relative capacitance variation of capacitive sensor based on the optimized hybrid dielectric. b) Detection of tiny pressure variations on the sensor that has been loaded with pressure of 1, 10, 100, and 450 kPa. c) Long-term stability of the sensor under a high pressure of 300 kPa for over 5000 cycles. d) Reproducibility of the high sensitivity and ultrawide linearity across different samples. e) Comparison among our work and other reported literatures. f) Output voltage variation of triboelectric tactile sensors based on the optimized hybrid dielectric.

capability of the sensor for ultralight weight detections. With the excellent deformability of MCA, the sensor can respond to the ultralight weights (e.g., a rice of ≈20 mg that is equal to ≈2 Pa) with the rapid response and relaxation time of ≈200 and 300 ms, respectively (Figure S19, Supporting Information). Additionally, we also noticed that with the MCA, the sensor possesses the relatively lower dependence on the precise sensor assembly (Figure S20, Supporting Information). Under different assembly conditions (e.g., with an air gap of ≈200 µm between the top electrode and hybrid dielectric, critical contact, and with a pressure loading of ≈50 Pa at initial state), the sensor based on the hybrid dielectric exhibited the much lower RSD of sensitivity (5.4%) than the one based on individual high-*k* RDA (39.7%). Figure S21 in the Supporting Information presents the dynamic monitoring of loading/withdrawing performance of the proposed sensor, where we can observe the repeated curves without obvious hysteresis thanks to the excellent elastic behavior of the dielectric structure. The dynamic relative capacitance variation of the sensor under different pressure intensities (e.g., 0.05, 0.5, 5, 50, and 500 kPa) was also examined (Figure S22, Supporting Information). The repeated square waves under each pressure intensity prove the excellent stability of the sensor. The capability of the sensor for long-term use was also confirmed by the periodic loading/unloading test (under a high pressure of 300 kPa for over 5000 cycles), as shown in Figure 3c. The stable and repeated signal reveals the stability of the sensor with the potential toward long-term applications. Figure S23 in the Supporting Information also provides the SEM images of the MCA after the fatigue test, where the MCA was slightly destroyed due to the high aspect ratio. Such a situation can be addressed by further optimizing the stability of MCA with higher mass ratio of PDMS gel and curing agent, longer curing time, and/or reducing the aspect ratio of MCA with lower magnetic field intensity, magnetic particle contents, etc. As revealed in Figure 3d, both the high sensitivity and ultrawide linearity range can be repeated across different samples, which indicates the excellent reproducibility of the proposed sensor devices and methodologies. We also compared the sensing performance of our proposed sensor with other reported literatures, as shown in Figure 3e. To our best knowledge, the capacitive pressure sensor that can exhibit

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such a high sensitivity in such an ultrawide linearity range is first reported.[7,8,11,14,17,23–25,35–41] It is worthwhile to note that the sensitivity of our sensor can be further enhanced while preserving the ultrawide linearity range by improving the intrinsic dielectric constant of the RDA components. Attributed to the unique series-parallel conversion, the initial capacitance of the sensor would not be obviously affected by regulating the intrinsic dielectric constant of the RDA components. However, the resultant capacitance of the sensor can be dramatically enhanced with the increasing dielectric constant, which can further improve the sensitivity. Without changing the structure, both the linear elastic modulus and linear dielectric behavior of the hybrid dielectric and hence the linearity of the sensor can be preserved.

In addition to the capacitive tactile sensors, we also demonstrated that our design of hybrid dielectric is applicable for triboelectric tactile sensors, as shown in Figure 3f. Based on the optimized MCA/RDA_{opt} hybrid dielectric, the sensor exhibited the similar ultrawide linearity range up to 1000 kPa (R^2 = 0.99) with a sensitivity of 0.196 mV kPa^{-1} (the detailed output voltage spectrum can be found in Figure S24 in the Supporting Information). To our best knowledge, the triboelectric tactile sensor that can exhibit such an ultrawide linearity range is rarely repo rted.[9,26–29,32,42–46] The linear voltage variation over the full scale of pressure spectrum can also be attributed to the linear effective dielectric constant of the optimized hybrid dielectric, which contributes to the linearly increased triboelectric charge density between the hybrid dielectric and electrodes under pressures (the detailed mechanism and discussions can be found in Figures S25 and S26 in the Supporting Information). Besides the ultrawide linearity range, we also noticed that the design of hybrid dielectric with combined low-*k* MCA and high-*k* RDA played a role in enhancing the triboelectrification and electrostatic induction for output voltage amplification (hence sensitivity enhancement). The voltage was obviously decreased when reducing the dielectric constant of RDA components or removing the low-*k* MCA of hybrid dielectrics, as shown in Figure S27 in the Supporting Information. The decrease of dielectric constant would debase the triboelectric charge density (i.e., the effect of triboelectrification), while the absence of MCA could somehow reduce the gap distance (i.e., the effect of electrostatic induction); they both resulted in the decrease of output voltage (the detailed discussion can be referred to Figure S27 in the Supporting Information). Note that in this work, the triboelectrification occurred with the contact between electrodes and dielectric, which somehow restricted the triboelectric charge density and hence the output voltage. By introducing other friction materials (e.g., poly(vinylidene fluoride), polyethylene terephthalate, etc.), the output voltage and hence sensitivity of the sensor can possibly be further improved.[47,48] The current study reveals that our design of hybrid dielectric is positive for the linearity across an ultrabroad pressure range of triboelectric tactile sensors, while future work will focus on the further optimization on the sensitivity.

2.4. Potential Applications as Physiological Signal Monitor and Command Controller

The skin, assisted by the nervous system that can transmit the induced biopotentials, functions with both tactile stimuli perceptions and feedback command collections (**Figure 4**a). Similar to the skin, the hybrid dielectric based sensors also exhibit the potential functions such as physiological signals sensing and control commands transmitting. Figure 4b presents the real-time response to the artery pulse when the capacitive sensor was attached to the wrist, where the pulse signals can be well repeated with an average frequency of ≈78 beats per minute in accordance with the value of normal adults. The typical pulse waveforms such as percussion, tidal, and diastolic peaks can also be recognized, which indicates the potential of the sensor for the healthcare monitoring and disease diagnosis. Here we did not specifically examine the triboelectric tactile sensor because the peak voltage signals were not clearly distinguished from the baseline due to the tiny pressure of artery pulses. Figure 4c presents the dynamic response to the breath when the device was attached inside a face mask. Both the capacitance and voltage signals in accordance with the respiratory rate can be recognized, which indicates the potential of the device for respiratory status monitoring. The sensor device can also detect the bending of knee joint when attached to the knee, as shown in Figure 4d. The stepwise signals with the increasing bending degrees and the well-repeated signals under the certain bending degree reveal that both the capacitive and triboelectric tactile sensor can serve as a body motion recorder. We also evaluated the detection capability of the capacitive/triboelectric sensor under higher pressure scope when the sensor device was attached to the foot, as shown in Figure 4e. The periodical waveforms can be stably observed under both the walking and running status. The status switch between walking and running can also be perceived in the form of obviously varied intensity and frequency of the capacitance/voltage signal. These results indicate that the capacitive/triboelectric tactile sensor can serve as not only a pedometer but also an action monitor thanks to the sensing capability with a broad pressure spectrum.

Beside the detection capability, we also demonstrated the capacitive/triboelectric sensor as a command transmitter, as shown in Figure 4f,g. Conventionally, flexible sensors always lack the capability to perceive the force with broad linearity, which imposes the difficulty to provide distinct signals that can precisely reflect the force intensities with a wide spectrum. Therefore, the special design of electronic interfaces and/or the combination of sensor pixels are typically required to transmit the multiple control commands.[49,50] Attributed to the high sensitivity and wide linearity range, our proposed tactile sensor can respond to the pressure variation with distinguishable signals across the ultrabroad pressure spectrum. The code groups as control instructions can thus be converted from a series of signal groups which is generated by combinatorial force inputs on a single sensor (Figure 4f). With predefined instructions, the mechanical inputs to the sensor device can then be used to control various electronic products, e.g., music players, and robots. Note that the control instructions can be generated by a single device, thus avoiding the high demand of numerous pixels integrated into an array and the complex signal processing and conversion. As a proof-of-concept, we herein demonstrated the 3-bit coded control instructions with the capacitive/triboelectric sensor, as shown in Figure 4g. The instruction codes such as "111," "222," "333," "123," "121," "122," and "321" were presented. For the capacitive sensor, the code numbers "1," "2,"

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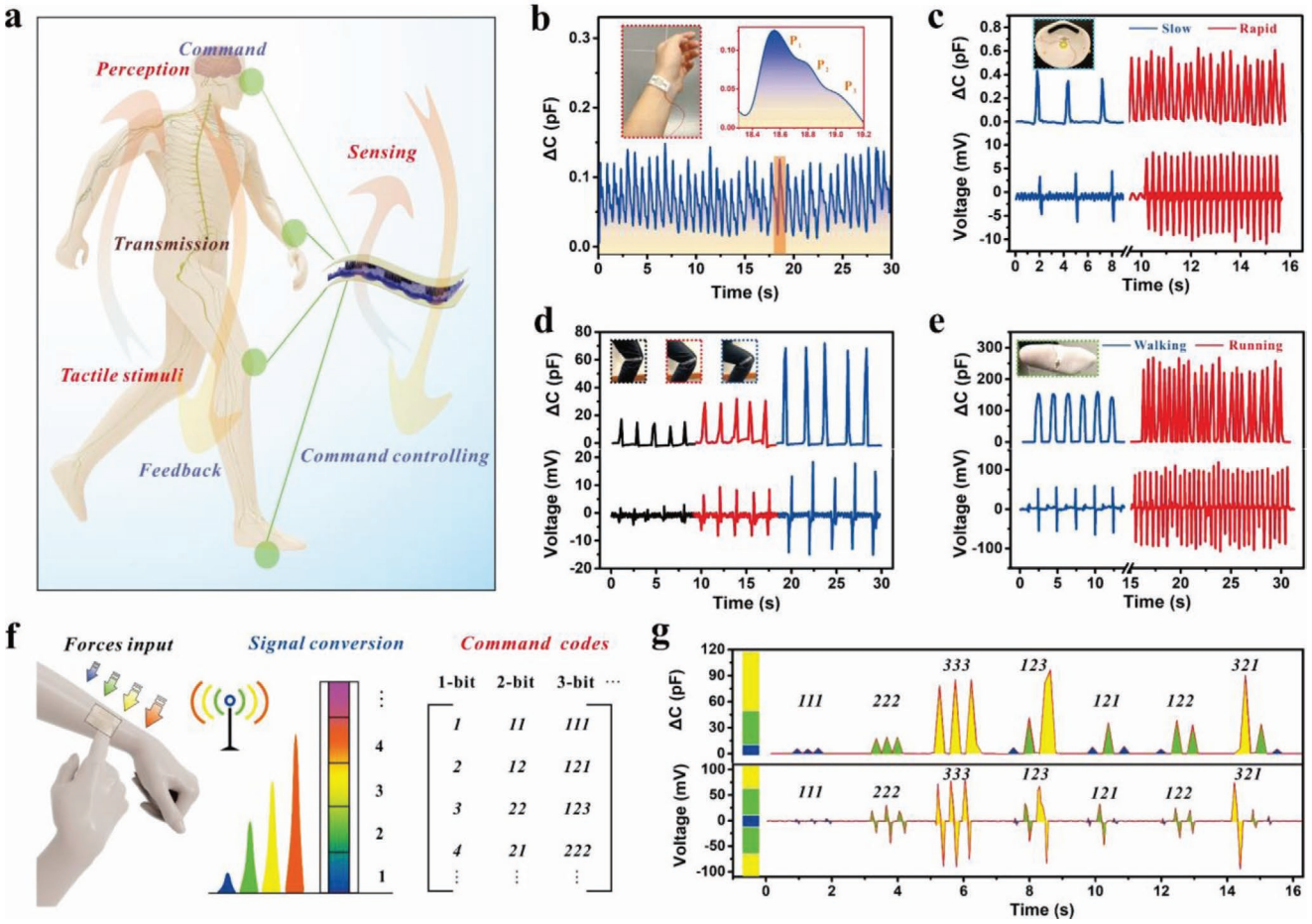

Figure 4. a) Multifunction of the skin and the hybrid dielectric based sensor. b) Physiological signals detection of artery pulse, c) respiratory rate, d) knee flexions, and e) walking and running. f) Schematic illustration of the command controller. g) 3-Bit coded control instructions output by both the capacitive and triboelectric tactile sensor.

and "3" were defined with the capacitance variations below 10, in the range of 10–50, and over 50 pF, respectively. For the triboelectric sensor, the code numbers were defined with the peak voltages below 10, in the range of 10–60, and over 60 mV, respectively. The intensities of capacitance and voltage signal were controlled by the finger press with low forces, medium forces and high forces thanks to the high pressure-resolution across the broad range. Based on the 3-bit codes, there are 27 control instructions in total that can be achieved by one single sensor device. In fact, taking advantages of the high-pressure resolution across the ultrawide pressure spectrum, we can expect that our sensors are also capable to output multi-bit codes with extensive control instructions, which can extremely simplify the control system integration, signal processing and conversion, etc.

3. Conclusion

We presented a novel strategy for the simultaneous optimization on linearity and sensitivity of capacitive tactile sensors based on the design of skin-inspired hybrid dielectric. With the combined low-k MCA and high-k RDA, the hybrid dielectric exhibited the gradient compressibility, gradient dielectric property, and pressure-induced series-parallel conversion, which enabled the separate customization of mechanical and dielectric behaviors. The gradient dielectric property with pressure-induced series-parallel conversion enabled the linearly increased effective dielectric constant and flexibly regulated initial/resultant capacitance based on the optimized hybrid dielectric. The gradient compressibility contributed to the linear behavior of elastic modulus with pressures regardless of the hybrid components regulations. Based on the linear elastic modulus, the linearity of capacitance variation would be mainly determined by the dielectric constant variation. As a result, the high sensitivity of 0.314 kPa^{-1} and the ultrawide linearity range up to 1000 kPa were simultaneously realized for capacitive tactile sensors, which is first reported. With the excellent sensitivity and linearity, the proposed sensor could thus preserve the high-pressure resolution over a wide pressure range. We also demonstrated that the hybrid dielectric was applicable for triboelectric tactile sensors. The combined low-k MCA and high-k RDA with linear effective dielectric constant not only enhanced the triboelectrification and electrostatic induction for voltage amplification but also realized the approximately linear variation in output voltage. Therefore, the optimized triboelectric sensor also exhibited the ultrawide linearity range (up to

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1000 kPa) with obviously enhanced sensitivity. As a proof-of-concept, both the proposed capacitive and triboelectric tactile sensor were successfully demonstrated as a healthcare monitor to detect the physiological signals such as the artery pulse, respiratory rate, joint flexion, walking and running status, etc. The proposed sensors were also applied as a command controller to provide the 3-bit coded control instructions. Attributed to the high-pressure resolution across the ultrawide pressure spectrum, the sensors are capable to generate different signal code groups based on the combination of exerted forces in a single device. With the dramatically improved sensing performance of the sensor, we believe that the novel design of hybrid dielectric can be promising for the flexible tactile sensors toward broad applications in the near future.

4. Experimental Section

Fabrication of CIP/NdFeB/PDMS Magnetic Composite and CNT/PDMS Mixture: For the fabrication of CIP/NdFeB/PDMS magnetic composite, the CIP (2 g; carbonyl iron (≥97% Fe basis), Sigma-Aldrich Chemistry, USA) and NdFeB particles (3 g; Magnequench, China) were first uniformly dispersed into cyclohexane solution (25 mL; Aladdin, China) with mechanical shaking and ultrasonic treatment (10 min). Then the PDMS gel (3 g; Sylgard 184 silicone elastomer kit, Dow Corning, USA) and curing agent (1 g) were added into the CIP/NdFeB/cyclohexane solution by the similar process to ensure the uniform distribution. The as-prepared magnetic composite was kept shaking to avoid sediment before use. For the preparation of CNT/PDMS mixture, the CNT (National Technology, China) and PDMS gel with certain mass ratio were dispersed into cyclohexane solution, respectively, and then were mixed together after the ultrasonic treatment (10 min). The CNT/PDMS/cyclohexane solution was stirred by a magnetic stirrer at 60 °C until the complete evaporation of cyclohexane. After uniformly mixing with the curing agent (mass ratio of 1:10 with employed PDMS gel), the CNT/PDMS mixture was obtained.

Fabrication Process of Hybrid Dielectric: The as-prepared CNT/PDMS mixture was firstly deposited into a plastic template with conversed micro-dome array in the squared groove, and compacted by a sandpaper with certain roughness. The fabrication of the plastic template can be referred to Figure S1 in the Supporting Information. After the curing process (80 °C, 6 h), removing of the sandpaper, and peeling off from the template, the RDA structure could be obtained. A through-hole PET mask was then attached to the rough surface of the as-prepared RDA structure, and the product was moved to the center of a permanent magnet (average surface magnetic field intensity of ≈400 mT) with a distance of ≈3 cm. The as-prepared CIP/NdFeB/PDMS magnetic composite was then sprayed onto the product (with the spraying distance of ≈20 cm) by an airbrush (with the injector diameter of 500 µm). The existence of the magnetic field, along with the effect of surface tension would thus lead to the automatic formation of micro-cilia structures as reported previously.[21] The NdFeB particles were employed to reduce the global magnetism of the CIP/NdFeB/PDMS composite for the relatively stable micro-cilia structure. After the solidification (50 °C, 12 h), the permanent magnet and the PET mask were removed, and the as-prepared hybrid structure was further cured at 100 °C for 2 h to ensure the stability of the micro cilia array. The detailed fabrication process and the regulation methodology of the hybrid dielectric can be found in Figure S1 in the Supporting Information. All the dielectrics were cut into 1 cm^2 for use. Based on the flexible PDMS, the hybrid dielectric also exhibits the excellent flexibility and bendability (Figure S28, Supporting Information). The hybrid dielectric and electrodes were assembled together for use assisted by a cilia-like fiber spacer with the excellent compressibility/elasticity and the similar thickness as the dielectric layer (Figure S28, Supporting Information). The Cu/Ni fabrics were employed as electrodes because of the excellent flexibility and conductivity (Figure S29, Supporting Information).

Characterizations and Measurements: The optical and SEM images were captured by Olympus optical microscopy and scanning electron microscopy (SEM, Sigma FE-SEM, Zeiss Corporation, Germany), respectively. The pressure loading and withdrawing were controlled by a motorized motion platform (Zolix, TSA50, China) with a motion controller (Zolix, MC600, China). The pressure was calculated by the ratio between the force value recorded by a precise balance (Sartorius Lab Instruments GmbH & Co. KG, 37 070 Goettingen, Germany) and the area between pressure loading and the sensing window (1 cm^2). The capacitance was monitored by the Keysight E4980A meter with a voltage supply of 1 V and a typical frequency of 1 kHz. The intrinsic dielectric constant and dielectric loss of CIP/NdFeB/PDMS and CNT/PDMS were measured by the same meter with a Keysight 16451B fixture. The effective dielectric constant of the hybrid dielectrics was calculated by $\varepsilon_{re} = Cd/A\varepsilon_0$ according to the definition of capacitance, where C is the measured capacitance under pressures, d the corresponding distance between the facing electrodes (i.e., dynamic height of the dielectric layer) and A the facing area between the electrodes (constant as 1 cm^2). The distance was measured from the captured high-resolution optical images of the sensor device deformed under different pressures. To accurately evaluate the mechanical and dielectric behavior of the hybrid dielectric, the dielectric layer and electrodes were fixed at the pressure loading/releasing platform without assembly for the measurement of mechanical deformation and effective dielectric constant. The elastic modulus of the hybrid dielectrics was calculated by $E = \frac{P}{\Delta d/d_0}$ according to the definition of elastic modulus, where P is the applied pressure and $\Delta d/d_0$ is the corresponding relative distance variation. The output voltage was captured by a Keysight 34972A digital multimeter with a fixed internal resistance of 10 MΩ. The frequency of pressure loading/withdrawing for triboelectric tactile sensors was fixed at ≈0.15 Hz due to the restriction of the experimental platform. The wearable demonstrations were carried out by a research volunteer with ensured safety during the characterization. Informed consents were obtained from the volunteer for the related tests.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

hybrid dielectrics, series-parallel conversion, tactile sensors, ultrawide linear sensing

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[1] D. Zhao, Y. Zhu, W. Cheng, W. Chen, Y. Wu, H. Yu, *Adv. Mater.* **2020**, https://doi.org/10.1002/adma.202000619.
[2] S. Yao, P. Ren, R. Song, Y. Liu, Q. Huang, J. Dong, B. T. O'Connor, Y. Zhu, *Adv. Mater.* **2020**, *32*, 1902343.
[3] O. A. Araromi, M. A. Graule, K. L. Dorsey, S. Castellanos, J. R. Foster, W. H. Hsu, A. E. Passy, J. J. Vlassak, J. C. Weaver, C. J. Walsh, R. J. Wood, *Nature* **2020**, *587*, 219.
[4] K. He, Y. Liu, M. Wang, G. Chen, Y. Jiang, J. Yu, C. Wan, D. Qi, M. Xiao, W. R. Leow, H. Yang, M. Antonietti, X. Chen, *Adv. Mater.* **2020**, *32*, 1905399.
[5] W. Jayathilaka, K. Qi, Y. Qin, A. Chinnappan, W. Serrano-Garcia, C. Baskar, H. Wang, J. He, S. Cui, S. W. Thomas, S. Ramakrishna, *Adv. Mater.* **2019**, *31*, 1805921.
[6] B. Ji, Y. Mao, Q. Zhou, J. Zhou, G. Chen, Y. Gao, Y. Tian, W. Wen, B. Zhou, *ACS Appl. Mater. Interfaces* **2019**, *11*, 28060.
[7] J. Wang, R. Suzuki, M. Shao, F. Gillot, S. Shiratori, *ACS Appl. Mater. Interfaces* **2019**, *11*, 11928.
[8] R. Y. Tay, H. Li, J. Lin, H. Wang, J. S. K. Lim, S. Chen, W. L. Leong, S. H. Tsang, E. H. T. Teo, *Adv. Funct. Mater.* **2020**, *30*, 1909604.
[9] M. Ha, S. Lim, S. Cho, Y. Lee, S. Na, C. Baig, H. Ko, *ACS Nano* **2018**, *12*, 3964.
[10] B. Ji, Q. Zhou, J. Wu, Y. Gao, W. Wen, B. Zhou, *ACS Appl. Mater. Interfaces* **2020**, *12*, 31021.
[11] Y. Xiong, Y. Shen, L. Tian, Y. Hu, P. Zhu, R. Sun, C.-P. Wong, *Nano Energy* **2020**, *70*, 104436.
[12] N. Bai, L. Wang, Q. Wang, J. Deng, Y. Wang, P. Lu, J. Huang, G. Li, Y. Zhang, J. Yang, K. Xie, X. Zhao, C. F. Guo, *Nat. Commun.* **2020**, *11*, 209.
[13] C. Jeong, J. S. Lee, B. Park, C. S. Hong, J. U. Kim, T. I. Kim, *Adv. Mater.* **2019**, *31*, 1902689.
[14] D. Kwon, T. I. Lee, J. Shim, S. Ryu, M. S. Kim, S. Kim, T. S. Kim, I. Park, *ACS Appl. Mater. Interfaces* **2016**, *8*, 16922.
[15] S. R. A. Ruth, V. R. Feig, H. Tran, Z. Bao, *Adv. Funct. Mater.* **2020**, *30*, 2003491.
[16] J. C. Yang, J. Mun, S. Y. Kwon, S. Park, Z. Bao, S. Park, *Adv. Mater.* **2019**, *31*, 1904765.
[17] M.-J. Yin, Z. Yin, Y. Zhang, Q. Zheng, A. P. Zhang, *Nano Energy* **2019**, *58*, 96.
[18] S. R. A. Ruth, L. Beker, H. Tran, V. R. Feig, N. Matsuhisa, Z. Bao, *Adv. Funct. Mater.* **2019**, *30*, 1903100.
[19] Q. Zhou, B. Ji, B. Hu, S. Li, Y. Xu, Y. Gao, W. Wen, J. Zhou, B. Zhou, *Nano Energy* **2020**, *78*, 105382.
[20] B. Ji, Q. Zhou, G. Chen, Z. Dai, S. Li, Y. Xu, Y. Gao, W. Wen, B. Zhou, *J. Mater. Chem. C* **2020**, *8*, 15634.
[21] Q. Zhou, B. Ji, Y. Wei, B. Hu, Y. Gao, Q. Xu, J. Zhou, B. Zhou, *J. Mater. Chem. A* **2019**, *7*, 27334.
[22] S. Park, H. Kim, M. Vosgueritchian, S. Cheon, H. Kim, J. H. Koo, T. R. Kim, S. Lee, G. Schwartz, H. Chang, Z. Bao, *Adv. Mater.* **2014**, *26*, 7324.
[23] A. Chhetry, S. Sharma, H. Yoon, S. Ko, J. Y. Park, *Adv. Funct. Mater.* **2020**, *30*, 1910020.
[24] J. Choi, D. Kwon, K. Kim, J. Park, D. D. Orbe, J. Gu, J. Ahn, I. Cho, Y. Jeong, Y. Oh, I. Park, *ACS Appl. Mater. Interfaces* **2020**, *12*, 1698.
[25] S. Sharma, A. Chhetry, M. Sharifuzzaman, H. Yoon, J. Y. Park, *ACS Appl. Mater. Interfaces* **2020**, *12*, 22212.
[26] S. Chen, N. Wu, S. Lin, J. Duan, Z. Xu, Y. Pan, H. Zhang, Z. Xu, L. Huang, B. Hu, J. Zhou, *Nano Energy* **2020**, *70*, 104460.
[27] J. Yu, X. Hou, J. He, M. Cui, C. Wang, W. Geng, J. Mu, B. Han, X. Chou, *Nano Energy* **2020**, *69*, 104437.
[28] Y. Jiang, K. Dong, X. Li, J. An, D. Wu, X. Peng, J. Yi, C. Ning, R. Cheng, P. Yu, Z. L. Wang, *Adv. Funct. Mater.* **2020**, *31*, 2005584.
[29] P. Zhang, Y. Chen, Z. H. Guo, W. Guo, X. Pu, Z. L. Wang, *Adv. Funct. Mater.* **2020**, *30*, 1909252.
[30] K. Dong, X. Peng, Z. L. Wang, *Adv. Mater.* **2020**, *32*, 1902549.
[31] H. Chen, Y. Song, X. Cheng, H. Zhang, *Nano Energy* **2019**, *56*, 252.
[32] Z. Zhao, Q. Huang, C. Yan, Y. Liu, X. Zeng, X. Wei, Y. Hu, Z. Zheng, *Nano Energy* **2020**, *70*, 104528.
[33] G. Zhao, Y. Zhang, N. Shi, Z. Liu, X. Zhang, M. Wu, C. Pan, H. Liu, L. Li, Z. L. Wang, *Nano Energy* **2019**, *59*, 302.
[34] H. R. Lim, H. S. Kim, R. Qazi, Y. T. Kwon, J. W. Jeong, W. H. Yeo, *Adv. Mater.* **2020**, *32*, 1901924.
[35] Q. Liu, Z. Liu, C. Li, K. Xie, P. Zhu, B. Shao, J. Zhang, J. Yang, J. Zhang, Q. Wang, C. F. Guo, *Adv. Sci.* **2020**, *7*, 2000348.
[36] H. Zhou, M. Wang, X. Jin, H. Liu, J. Lai, H. Du, W. Chen, A. Ma, *ACS Appl. Mater. Interfaces* **2021**, *13*, 1441.
[37] S. Sharma, A. Chhetry, S. Zhang, H. Yoon, C. Park, H. Kim, Md. Sharifuzzaman, X. Hui, J. Y. Park, *ACS Nano* **2021**, *15*, 4380.
[38] Y. Luo, J. Shao, S. Chen, X. Chen, H. Tian, X. Li, L. Wang, D. Wang, B. Lu, *ACS Appl. Mater. Interfaces* **2019**, *11*, 17796.
[39] J. Yang, S. Luo, X. Zhou, J. Li, J. Fu, W. Yang, D. Wei, *ACS Appl. Mater. Interfaces* **2019**, *11*, 14997.
[40] T. Jin, Y. Pan, G. J. Jeon, H. I. Yeom, S. Zhang, K. W. Paik, S. K. Park, *ACS Appl. Mater. Interfaces* **2020**, *12*, 13348.
[41] J. Qiu, X. Guo, R. Chu, S. Wang, W. Zeng, L. Qu, Y. Zhao, F. Yan, G. Xing, *ACS Appl. Mater. Interfaces* **2019**, *11*, 40716.
[42] E. Kar, N. Bose, B. Dutta, N. Mukherjee, S. Mukherjee, *ACS Appl. Mater. Interfaces* **2019**, *11*, 17501.
[43] Y. Park, Y. E. Shin, J. Park, Y. Lee, M. P. Kim, Y. R. Kim, S. Na, S. K. Ghosh, H. Ko, *ACS Nano* **2020**, *14*, 7101.
[44] P. Maharjan, T. Bhatta, M. Salauddin, M. S. Rasel, M. T. Rahman, S. M. S. Rana, J. Y. Park, *Nano Energy* **2020**, *76*, 105071.
[45] T. An, D. V. Anaya, S. Gong, L. W. Yap, F. Lin, R. Wang, M. R. Yuce, W. Cheng, *Nano Energy* **2020**, *77*, 105295.
[46] Y.-W. Cai, X.-N. Zhang, G.-G. Wang, G.-Z. Li, D.-Q. Zhao, N. Sun, F. Li, H.-Y. Zhang, J.-C. Han, Y. Yang, *Nano Energy* **2021**, *81*, 105663.
[47] H. Zou, Y. Zhang, L. Guo, P. Wang, X. He, G. Dai, H. Zheng, C. Chen, A. C. Wang, C. Xu, Z. L. Wang, *Nat. Commun.* **2019**, *10*, 1427.
[48] N. Gogurla, B. Roy, J.-Y. Park, S. Kim, *Nano Energy* **2019**, *62*, 674.
[49] Q. Shi, C. Lee, *Adv. Sci.* **2019**, *6*, 1900617.
[50] G. H. Lee, J. K. Park, J. Byun, J. C. Yang, S. Y. Kwon, C. Kim, C. Jang, J. Y. Sim, J. G. Yook, S. Park, *Adv. Mater.* **2020**, *32*, 1906269.

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